

**Constraint-Based Pricing in Cloud Computing: Evidence from AWS Spot
Instance Regional Price Dispersion**

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Date: June 26, 2026

Version: V3.0

Keywords: constraint-based pricing, cloud computing, spot instances, infrastructure-defined assets, cross-domain validation

AI Disclosure: The core research question, theoretical framework, and all principal arguments of this paper were independently proposed by the author. During the writing process, the author engaged in dialogue with an AI tool (DeepSeek) for clarifying and refining argumentation logic, and for language polishing. The AI did not participate in the generation of core ideas or the construction of the theoretical framework. The author bears full responsibility for the entire content of this paper.

Notation

Symbol	Meaning
P_{spot}	Spot instance price (\$/h)
$P_{on-demand}$	On-demand instance price (\$/h)
η	Constraint completeness = $1 - interruption_rate$
$ \Omega $	Physical server capacity (number of spare servers)
$f(capacity_utilization)$	Utilization function: higher utilization $\rightarrow$ higher Spot price

Abstract

Why does identical compute capacity — the same `m6i.large` virtual machine instance — trade at different Spot prices across AWS regions, when the on-demand price is uniform? Across five AWS regions, the Spot price for this instance type ranges from \$0.0253/h to \$0.0373/h, a spread of 47%. Standard explanations invoke "supply and demand," which restates the observation without revealing its structure. This paper argues that AWS Spot Instance pricing is a realized case of constraint-based asset pricing: the Spot price is a function of the constraint structure — interruption rate, capacity utilization, and the institutional rules governing access to spare compute capacity. Using publicly verifiable region-level pricing data, we show that the 47% price dispersion cannot be explained by differences in interruption rates (all five regions share the same <5% band). The residual variation is attributable to capacity utilization — the tightness of spare server capacity in each region. A Monte Carlo sensitivity analysis confirms the monotonic relationship between constraint completeness η and Spot price under model assumptions. The finding establishes cloud computing alongside carbon allowances and data access rights as domains where constraint-based pricing identifies structural differentiation masked by uniform or semi-uniform pricing. A complete data processing specification is provided for future availability zone-level validation using the AWS Spot Price History public dataset (Zenodo DOI: 10.5281/zenodo.18447244).

1. Introduction

A standard `m6i.large` virtual machine — 2 vCPUs, 8 GB of RAM — costs \$0.096 per hour on-demand in every AWS region where it is available. The on-demand price is uniform. The Spot price is not.

In us-east-1 (Northern Virginia), the Spot price for this instance is \$0.0348/h. In us-west-2 (Oregon), it is \$0.0373/h — 7% higher. In mx-central-1 (Mexico), it is \$0.0253/h — 27% lower than Virginia. Across five regions, the Spot price for identical compute capacity varies by **47%** from lowest to highest, while the on-demand price remains fixed at \$0.096/h.

AWS attributes Spot pricing to "long-term supply and demand trends for spare EC2 capacity." This explanation is not incorrect, but it is structurally empty. Every price in every market is determined by supply and demand. The question is: what determines the supply and demand for spare server capacity in a specific AWS region? Why is that spare capacity systematically tighter in Oregon than in Mexico? Why is the price difference persistent rather than random?

This paper argues that AWS Spot Instance pricing is a realized case of **constraint-based asset pricing** (Mao, 2026b) — a framework in which the value of a resource is a function of the institutional and physical constraints that define access to it. In this framework, the Spot price is not an emergent market price. It is the output of a pricing function whose inputs are the constraint structure of each region: the interruption rate (how reliably the instance will not be reclaimed by AWS), the capacity utilization rate (how much spare capacity exists), and the institutional rules governing who may access it and under what conditions.

The contribution of this paper is threefold. First, it demonstrates that a major cloud provider's pricing algorithm implements — without naming it as such — the same constraint-based pricing logic that has been independently formalized for low-altitude airspace, carbon emission allowances, and data access rights. Second, it establishes cloud computing as a clean, publicly verifiable testbed for constraint-based pricing: the price data is publicly archived, and the constraint parameters (interruption rate bands,

instance availability) are partially observable through AWS's own tools. Third, it provides a complete specification for availability zone-level validation, enabling subsequent researchers to reproduce and extend the analysis.

2. The Constraint Structure of AWS Spot Instances

2.1 Spot Instances as Constraint-Defined Assets

A Spot instance is not a guaranteed virtual machine. It is the right to use a virtual machine *conditional on AWS not needing the underlying physical server for on-demand or reserved instance customers*. When AWS needs the capacity back, it reclaims the Spot instance with a two-minute warning. The access right is interruptible.

This conditional structure defines a constraint-based asset in the precise sense of the framework: its value depends not on the physical properties of the compute capacity (identical across instances of the same type), but on the institutional rules governing who gets to use it, under what conditions, and how reliably those conditions are enforced.

2.2 Four-Layer Constraint Architecture

The constraint structure of a Spot instance can be decomposed into four layers:

Layer	Constraint Type	AWS Implementation	Key Parameter
L ₁ Physical	Physical server capacity per data center	Limited by rack space, power, cooling	
L ₂ Digital Compliance	Authentication and authorization	AWS API, IAM roles, service quotas	Access control integrity

Layer	Constraint Type	AWS Implementation	Key Parameter
L ₃ Financial	Pricing mechanism for interruptible capacity	Spot market — price varies by region and instance type	P _{spot} vs P _{on-demand}
L ₄ Governance	Rules for interruption and reclamation	2-minute warning; instance can be terminated, stopped, or hibernated	η : 1 – interruption_rate

The key parameters for pricing are:

- **$\eta = 1 - \text{interruption_rate}$** : Constraint completeness. Measures how reliably the Spot instance will remain available. A region where AWS rarely reclaims Spot capacity has η close to 1. A region with frequent interruptions has lower η .
- **Capacity utilization**: The fraction of physical servers of a given type that are currently in use. When utilization is high, spare capacity is scarce, and Spot prices rise.

2.3 The Pricing Function

In the constraint-based pricing framework (Mao, 2026b), the value of a constraint-defined asset is:

$$V = N \times \text{Cred} \times H \times |\Omega| \times \bar{P} \times \bar{q} / (1 + r_f + \text{RP})$$

For Spot instances, this simplifies. The negation channel exists ($N = 1$). The channel credibility Cred is approximately equal to η (the probability of non-interruption). The multi-layer completeness discount H simplifies to η in the single-layer analysis. $\bar{P}$ is the on-demand price — willingness to pay for guaranteed access. The resulting Spot

price expression is:

$$P_{\text{spot}} \approx P_{\text{on-demand}} \times \eta \times f(\text{capacity_utilization})$$

where f is a decreasing function: the higher the utilization, the higher the Spot price.

This yields two testable predictions:

1. **Within the same interruption rate band**, regions with tighter capacity should show higher Spot prices.
2. **Across different interruption rate bands**, regions with lower interruption rates (higher η) should show higher Spot prices, all else equal.

3. Empirical Analysis

3.1 Data

Region-Level Data. We use publicly available pricing data for `m6i.large` instances (2 vCPU, 8 GB RAM, Intel Xeon Ice Lake) across five AWS regions, sourced from aws-pricing.com and accessed on June 25, 2026. Interruption rate bands are sourced from the AWS Spot Instance Advisor (aws.amazon.com/ec2/spot/instance-advisor/), which reports bands of <5%, 5–10%, 10–15%, 15–20%, and >20%.

AWS Region	Spot Price (/h)	On — Demand (/h)	Interruption Band
mx-central-1 (Mexico)	\$0.0253	\$0.096	<5%
us-east-2 (Ohio)	\$0.0269	\$0.096	<5%
ap-south-1 (Mumbai)	\$0.0320	\$0.096	<5%
us-east-1 (N. Virginia)	\$0.0348	\$0.096	<5%
us-west-2 (Oregon)	\$0.0373	\$0.096	<5%

All five regions share the same on-demand price (\$0.096/h) and the same interruption rate band (<5%). The Spot price spread is 47% between the lowest and highest.

Availability Zone-Level Data. Full AZ-level analysis requires the AWS Spot Price History public dataset (Zenodo DOI: 10.5281/zenodo.18447244), which spans 2022–2026 at 5-minute granularity. The complete data processing specification is provided in Section 3.4. The current analysis is based on region-level data and is transparent about this limitation.

3.2 Constraint Structure Analysis

The 47% Spot price dispersion across five regions with identical on-demand pricing and identical interruption rate bands presents a clear signal: the interruption rate alone cannot explain the price variation. All five regions are in the <5% band, yet their Spot prices differ substantially. The residual driver must be **capacity utilization** — the ratio of spare servers to demand in each region.

us-west-2 (Oregon) commands the highest Spot price at \$0.0373/h — consistent with a mature, high-demand region with tight spare capacity. mx-central-1 (Mexico) has the lowest Spot price at \$0.0253/h — consistent with a newer region that AWS is filling with capacity ahead of demand. The rank ordering maps to intuitive expectations about regional cloud infrastructure maturity: newer regions have more headroom; established tech hub regions are tighter.

Testable prediction: If AWS were to publish per-region capacity utilization data (currently unavailable), the constraint pricing framework predicts that the rank order of utilization rates would match the rank order of Spot prices shown above.

3.3 Monte Carlo Sensitivity Analysis

To examine the η –Spot price relationship under model assumptions, we conduct a Monte Carlo simulation with 5,000 parameter draws. Parameters are sampled from uniform distributions: $\eta \in [0.85, 0.99]$, capacity utilization $\in [0.30, 0.90]$. **The simulation tests model sensitivity, not empirical validation:** it shows how Spot price responds to η *under the assumption that the constraint pricing relationship holds*.

η Range	Average Spot Price (Simulated)	Avg Savings vs On-Demand
[0.85, 0.90)	\$0.0403/h	58%
[0.90, 0.95)	\$0.0432/h	55%
[0.95, 1.00)	\$0.0452/h	53%

The monotonic relationship is confirmed in the model: higher η (more reliable access) raises the Spot price toward the on-demand price. The magnitude of the simulated effect (a few dollars per hour across a wide η range) is comparable to the observed regional price dispersion (a few cents per hour across all-<5% regions), suggesting the model's parameterization is qualitatively consistent with observed data.

3.4 Specification for Availability Zone-Level Validation

For rigorous testing at the AZ level, the following workflow is specified:

Data: AWS Spot Price History, Zenodo DOI: 10.5281/zenodo.18447244 (~4.2 GB, 2022–2026, monthly ZStandard-compressed TSV files). **Environment:** Python 3.10+ with `zstandard`, `pandas`, `scipy.stats`, `matplotlib`. **Pipeline:** (1) Filter to `m6i.large` in `us-east-1`. (2) Group by AZ; compute mean, SD, and time series of Spot prices. (3) Correlate with interruption bands via ANOVA or Kruskal-Wallis. (4)

For AZs in the same band, test pairwise price differences (t-test). **Criterion:** If AZs in the same interruption band show significantly different Spot prices ($p < 0.05$), capacity utilization is confirmed as a pricing factor at the AZ level.

4. Cross-Domain Validation

The AWS Spot finding extends the empirical base of constraint-based pricing beyond its original application domains. The same constraint structure framework identifies pricing differentiation — or its absence — across three independent markets:

Domain	Constraint Resource	Observed η Metric	Current Pricing Pattern
Cloud (AWS Spot)	Server capacity per region/AZ	Interruption rate band	Different regions, different prices — region-level confirmed
Carbon (CEA)	Emission allowances by industry	MRV audit quality	Same price across industries — prediction: should differentiate
Data API	Data access rights by buyer type	Access audit traceability	Same price for all buyers — prediction: should differentiate

In cloud computing, constraint-differentiated pricing is already in production at the region level. In carbon and data markets, the differentiation predicted by the framework has not yet materialized — presenting both a testable hypothesis and a potential arbitrage opportunity.

5. Conclusion

AWS Spot Instance pricing at the region level is consistent with the constraint-based pricing framework. Identical compute capacity trades at different prices across regions, the on-demand baseline is uniform, interruption rate bands are identical — and yet Spot prices vary by 47%. The residual driver, capacity utilization, is not directly observable but its predicted effects match the observed rank ordering across regions.

The current analysis is restricted to publicly verifiable region-level data. Availability zone-level validation — the natural next step — has been fully specified. The required dataset is publicly archived on Zenodo with a DOI. The processing pipeline, statistical tests, and decision criteria are documented. Any researcher can execute the next phase of validation without ambiguity.

Until AZ-level validation is complete, the contribution of this paper is: **region-level evidence supports the constraint pricing hypothesis; a precise AZ-level test has been formulated and awaits execution.** This transparency is not a limitation — it is the operational definition of reproducibility.

References

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Working paper V3.0. Region-level pricing data from aws-pricing.com, accessed June 25, 2026. Interruption rate bands from AWS Spot Instance Advisor. AZ-level dataset: Zenodo DOI [10.5281/zenodo.18447244](https://doi.org/10.5281/zenodo.18447244).